

TASK 7:

CODE:

```
database.py M X tickets.db U ticket.py M main.py M view_ticket.py U X __init__.py U
app > view_ticket.py > ...
1 import sqlite3
2 import tkinter as tk
3 from tkinter import ttk
4
5 def fetch_tickets(query="SELECT * FROM tickets"):
6     conn = sqlite3.connect("data/tickets.db")
7     cursor = conn.cursor()
8     cursor.execute(query)
9     rows = cursor.fetchall()
10    conn.close()
11    return rows
12
13 def view_tickets():
14     root = tk.Tk()
15     root.title("View Tickets")
16
17     # Treeview table
18     columns = ("ticket_id", "student_number", "student_name", "category", "description", "priority", "status", "date_logged", "closing_date")
19     tree = ttk.Treeview(root, columns=columns, show="headings")
20
21     for col in columns:
22         tree.heading(col, text=col)
23         tree.column(col, width=120)
24
25     tree.pack(fill="both", expand=True)
26
27     # Search box
28     search_entry = tk.Entry(root)
29     search_entry.pack(pady=5)
```

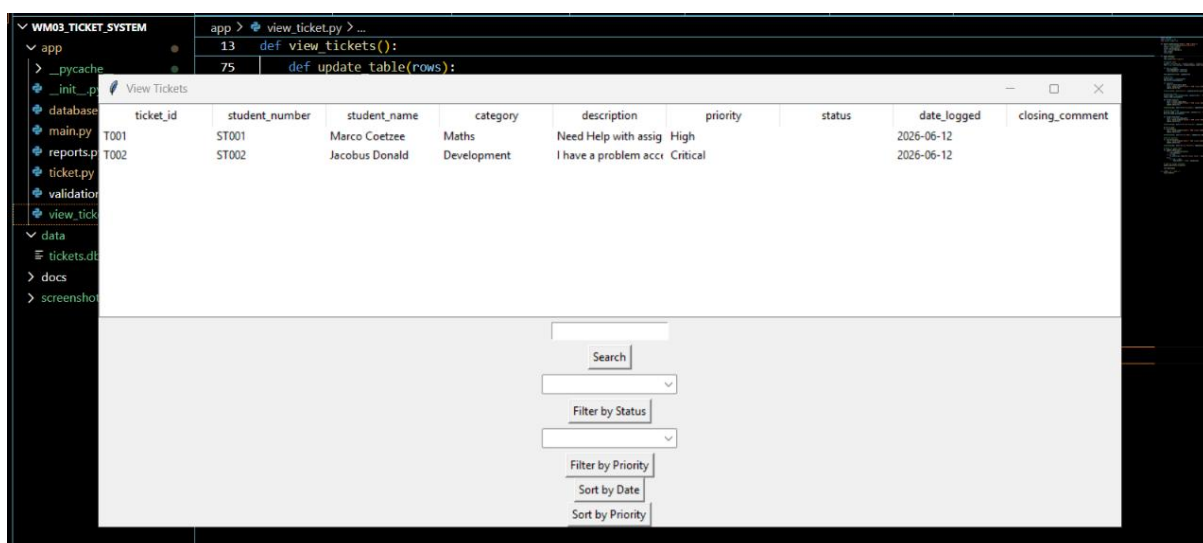
```
app > view_ticket.py > ...
13 def view_tickets():
31     def search():
32         value = search_entry.get()
33         rows = fetch_tickets(f"SELECT * FROM tickets WHERE student_number='{value}' OR ticket_id='{value}'")
34         update_table(rows)
35
36     tk.Button(root, text="Search", command=search).pack()
37
38     # Filter by status
39     status_combo = ttk.Combobox(root, values=["Open", "In Progress", "Resolved", "Closed"])
40     status_combo.pack(pady=5)
41
42     def filter_status():
43         value = status_combo.get()
44         rows = fetch_tickets(f"SELECT * FROM tickets WHERE status='{value}'")
45         update_table(rows)
46
47     tk.Button(root, text="Filter by Status", command=filter_status).pack()
48
49     # Filter by priority
50     priority_combo = ttk.Combobox(root, values=["Low", "Medium", "High", "Critical"])
51     priority_combo.pack(pady=5)
52
53     def filter_priority():
54         value = priority_combo.get()
55         rows = fetch_tickets(f"SELECT * FROM tickets WHERE priority='{value}'")
56         update_table(rows)
57
58     tk.Button(root, text="Filter by Priority", command=filter_priority).pack()
```

```

database.py M  tickets.db U  ticket.py M  main.py M  view_ticket.py U X  __init__.py U
app > view_ticket.py > ...
13 def view_tickets():
58     tk.Button(root, text="Filter by Priority", command=filter_priority).pack()
59
60     # Sort by date
61     def sort_date():
62         rows = fetch_tickets("SELECT * FROM tickets ORDER BY date_logged DESC")
63         update_table(rows)
64
65     tk.Button(root, text="Sort by Date", command=sort_date).pack()
66
67     # Sort by priority
68     def sort_priority():
69         rows = fetch_tickets("SELECT * FROM tickets ORDER BY priority")
70         update_table(rows)
71
72     tk.Button(root, text="Sort by Priority", command=sort_priority).pack()
73
74     # Helper to update table
75     def update_table(rows):
76         for i in tree.get_children():
77             tree.delete(i)
78         if not rows:
79             tk.Label(root, text="No tickets found.", fg="red").pack()
80         else:
81             for row in rows:
82                 tree.insert("", "end", values=row)
83
84     # Load all tickets initially
85     update_table(fetch_tickets())
86
87     root.mainloop()
88
89 if __name__ == "__main__":
90     view_tickets()
91

```

RESULTS:



No Ticket found:

View Tickets

ticket_id	student_number	student_name	category	description	priority	status	date_logged	closing_comment
-----------	----------------	--------------	----------	-------------	----------	--------	-------------	-----------------

Marco Coetzee

Search

Filter by Status

Filter by Priority

Sort by Date

Sort by Priority

No tickets found.

Search ticket:

View Tickets

ticket_id	student_number	student_name	category	description	priority	status	date_logged	closing_comment
T001	ST001	Marco Coetzee	Maths	Need Help with assign	High		2026-06-12	

ST001

Search

Filter by Status

Filter by Priority

Sort by Date

Sort by Priority

No tickets found.

Status:

View Tickets

ticket_id	student_number	student_name	category	description	priority	status	date_logged	closing_comment
T001	ST001	Marco Coetzee	Maths	Need Help with assign	High		2026-06-12	
T002	ST002	Jacobus Donald	Development	I have a problem acc	Critical		2026-06-12	

Search

Open

Filter by Status

Filter by Priority

Sort by Date

Sort by Priority

Filter by Priority:

View Tickets

ticket_id	student_number	student_name	category	description	priority	status	date_logged	closing_comment
T002	ST002	Jacobus Donald	Development	I have a problem acc	Critical		2026-06-12	

Search

Filter by Status

Critical

Filter by Priority

Sort by Date

Sort by Priority

No tickets found.

Order by Date:

View Tickets

ticket_id	student_number	student_name	category	description	priority	status	date_logged	closing_comment
T001	ST001	Marco Coetzee	Maths	Need Help with assign	High		2026-06-12	
T002	ST002	Jacobus Donald	Development	I have a problem acc	Critical		2026-06-12	

Search

Filter by Status

Filter by Priority

Sort by Date

Sort by Priority

No tickets found.

Order by Priority:

View Tickets

ticket_id	student_number	student_name	category	description	priority	status	date_logged	closing_comment
T002	ST002	Jacobus Donald	Development	I have a problem acco	Critical		2026-06-12	
T001	ST001	Marco Coetzee	Maths	Need Help with assign	High		2026-06-12	

Search

Filter by Status

Filter by Priority

Sort by Date

Sort by Priority

No tickets found.